Assignment 1

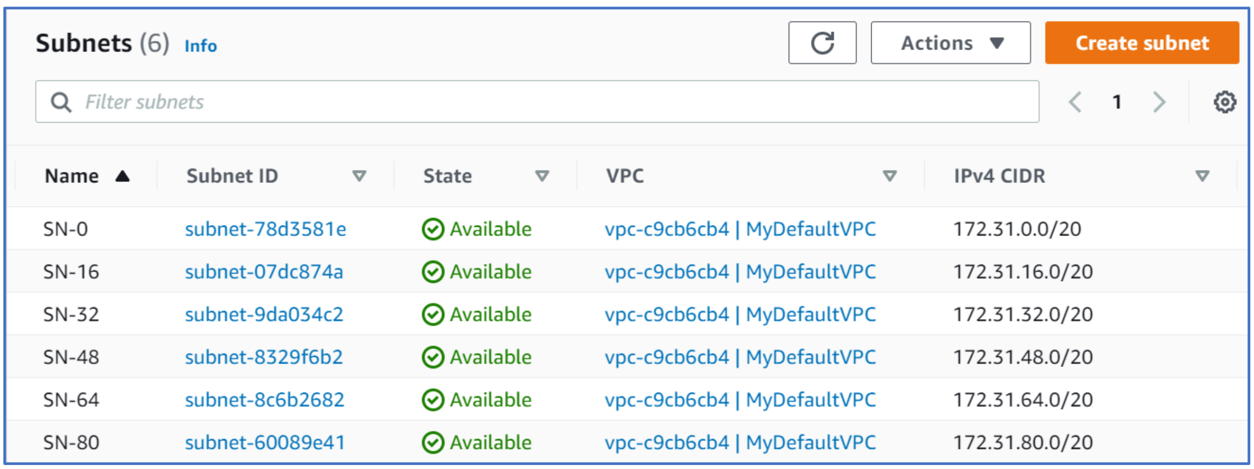
AWS Subnets, Network Access Control Lists and Security Groups

The objective of this assignment is to create a network infrastructure in the default VPC in the AWS account. This is to be accomplished by making **new subnets** and modifying the **network access control lists (NACL)** and **security groups**.

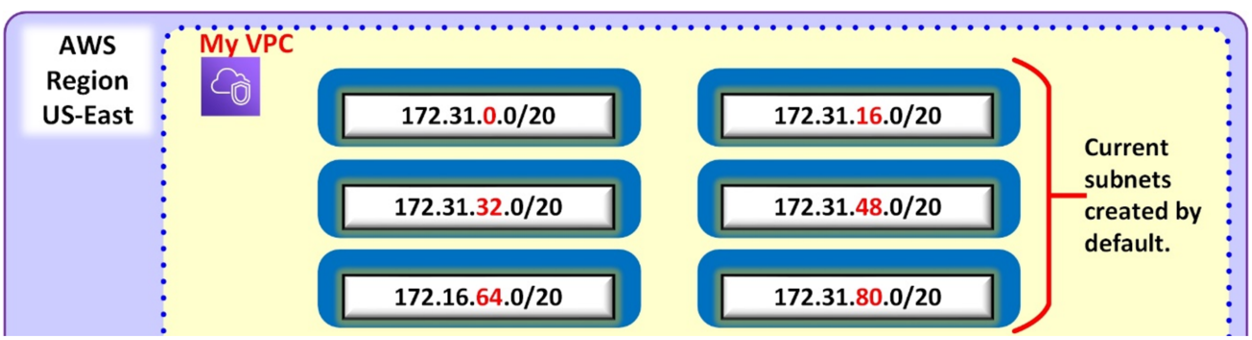
For all resources created in this assignment, you should have your name as part of the resource name. For example,

General Description

The default VPC already has these subnets in the address space 172.31.0.0/16:

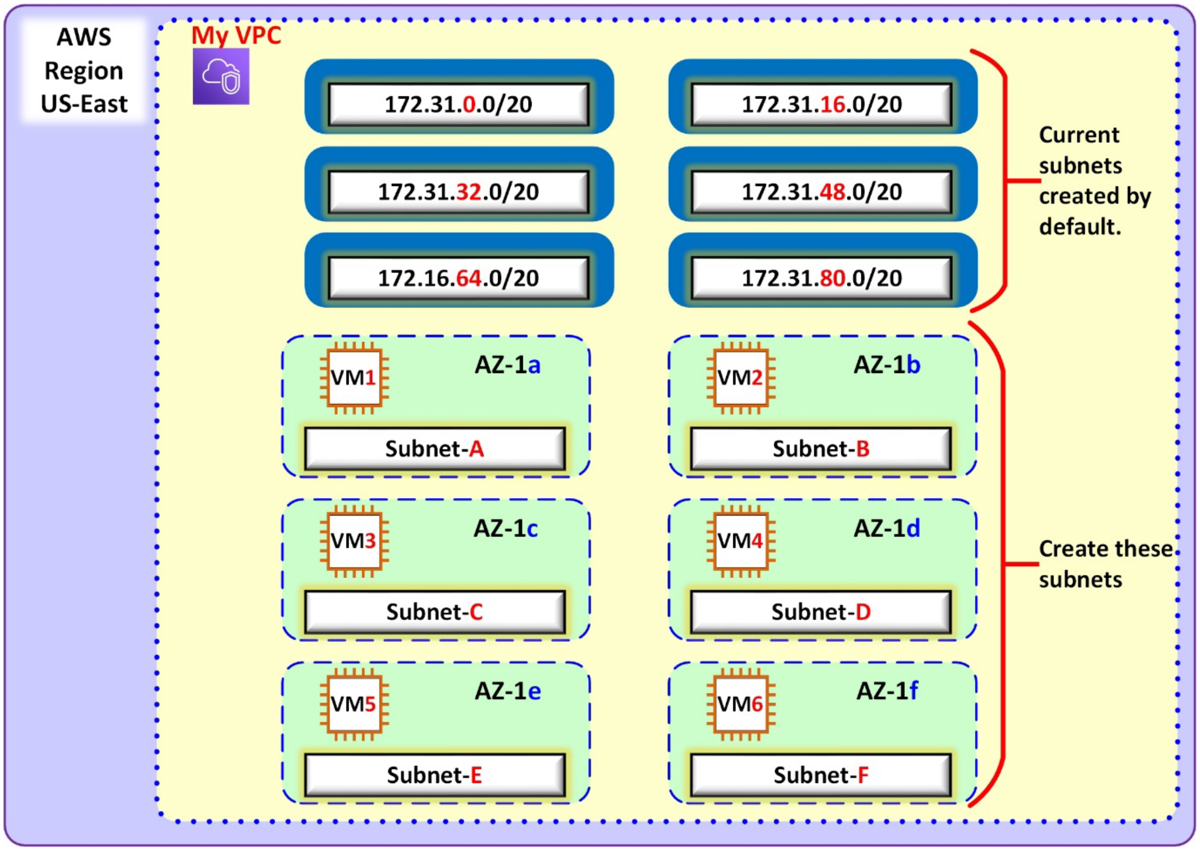

Subnets (6) Info

Name	Subnet ID	State	VPC	IPv4 CIDR
SN-0	subnet-78d3581e	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.0.0/20
SN-16	subnet-07dc874a	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.16.0/20
SN-32	subnet-9da034c2	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.32.0/20
SN-48	subnet-8329f6b2	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.48.0/20
SN-64	subnet-8c6b2682	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.64.0/20
SN-80	subnet-60089e41	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.80.0/20



- Each of these subnets has a mask /20 enough for $2^{(32-20)}$= 4096 addresses.
- The space 172.31.0.0/16 still have address space to create more subnets.
- In this project, you will do the **subnetting** according to the required network size to create six (6) more subnets.
- The **subnets** will be deployed in different **availability zones**.
- In every new subnet, **EC2-instances** will be deployed for testing connection.
- Every subnet will have a particular **NACL**.
- **Security Groups** will control the access to the EC2-intances.

This must be the VPC's topology at the end of this project:



Procedure

(Submissions of results in one report in Word or PDF).

TASK 1: Creation of subnets

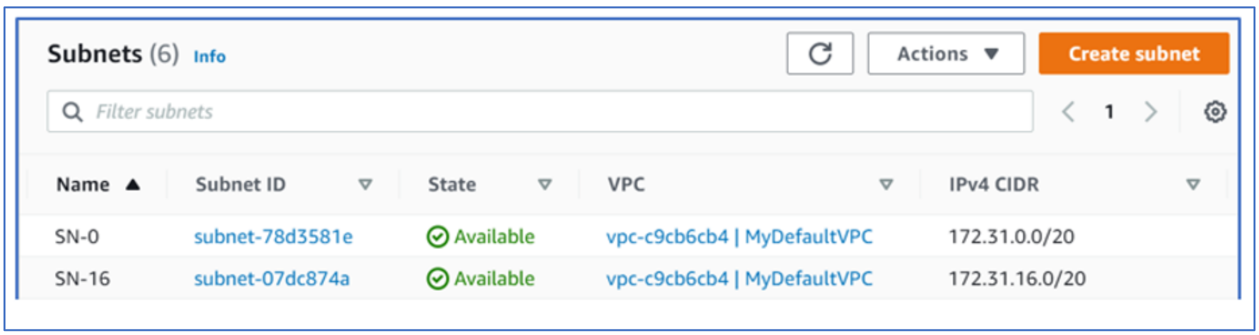
- Create six subnets within the address space 172.31.0.0/16 with the following characteristics. Make sure to include your name in the subnet name (example SN-A-saber).

Subnet name	Number of IPv4 addresses	Availability Zone
SN-A	2048	US-east-1a
SN-B	2048	US-east-1b
SN-C	4096	US-east-1c
SN-D	8192	US-east-1d
SN-E	8192	US-east-1e
SN-F	16,384	US-east-1f

- For example, subnet A (SN-A) will have 2,048 addresses in total including the subnet address, the broadcast address, and the reserved addresses and it will be created in the US-east region availability zone 1a. Follow the table to create your subnets.

SUBMIT for TASK 1:

- The snippets (follow each checkmark) from the AWS Subnet tab clearly showing:
  - ✓ Each **subnet**
  - ✓ The **subnet Names**.
  - ✓ The **IPV4 CIDR** of each subnet.
  - ✓ The **number of addresses** available of each subnet.
  - ✓ The **availability zone** of each subnet.
  - ✓ **Auto-assign public IPv4** address condition of each subnet.


Subnets (6) Info

Actions | Create subnet

Filter subnets

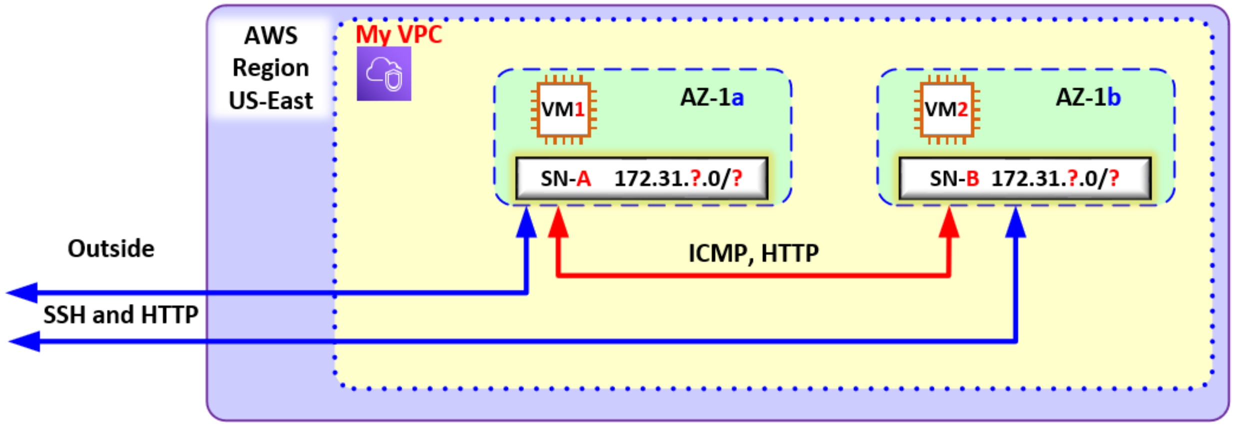
Name	Subnet ID	State	VPC	IPv4 CIDR
SN-0	subnet-78d3581e	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.0.0/20
SN-16	subnet-07dc874a	Available	vpc-c9cb6cb4 | MyDefaultVPC	172.31.16.0/20

TASK 2: Configuration of Access

- Configuration of Network Access Control NACL.
  - ✓ Configure one NACL for every Subnet.
- Configuration of Security Groups.
  - ✓ Create Security Groups as indicated in this table.
  - ✓ Make sure to include your name in the NACL and Security group names (example NACL-A-saber, webSG-saber)

Subnet name	NACL name	Security Group Name
SN-A	NACL-A	webSG
SN-B	NACL-B	webSG
SN-C	NACL-C	webSG
SN-D	NACL-D	WinSG
SN-E	NACL-E	WinSG
SN-F	NACL-F	webSG

- Associate the NACL to the respective subnet.
- Implement the following cases by customizing the **NACL**s or the **Security Group**s or both.
- **Note**: even though the EC2 instances are shown in the diagram, they are not to be deployed yet.

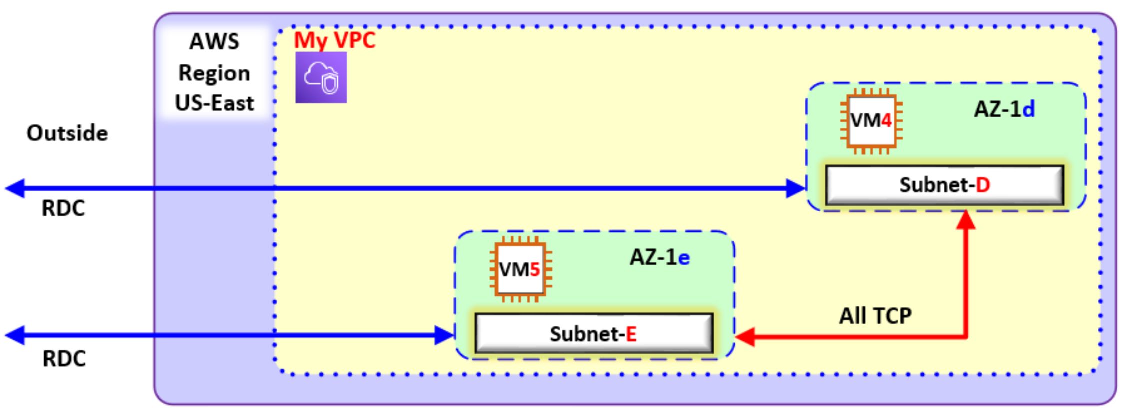


- The machine VM1 can be accessed from your home address via SSH and HTTP.
- The machine VM2 can be accessed from your home address via SSH and HTTP.
- The machines (any) attached to subnet SN-A can ping and HTTP any machine attached to SN-B and vice versa, any machine located in SN-B can ping and HTTP any machine attached to SN-A.
- The previous conditions are exclusive. No other traffic, from anywhere is allowed.
- **Note**: to test HTTP, **curl** to the private IPv4 address of the other machine.

SUBMIT for task 2a

✓ The snippets of your solution (whatever you used, Security Group or NACLs or a combination).
✓ The explanation of your solution.

- Implement the following case by customizing the **NACL**s or the **Security Group**s or both.

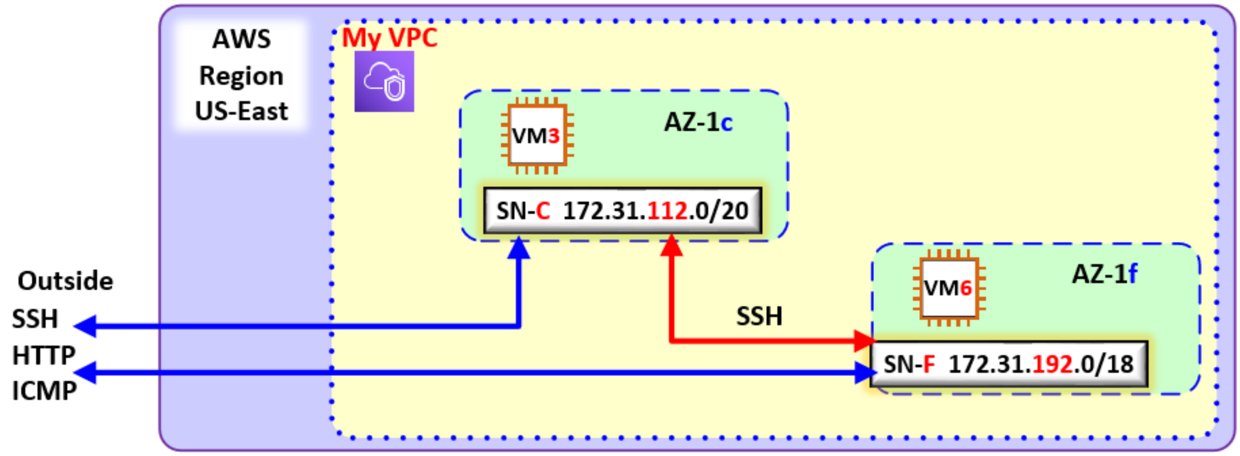


- Subnets D and E are for the Windows Server farm.
- The hosts connected to SN-E and SN-D can access each other with the protocols ICMP and TCP. SSH access from your home address is allowed for administration reasons.
- The hosts connected to SN-D and SN-E can access each other with SSH and HTTP. SSH access from your home address is allowed for administration reasons.
- The hosts connected to SN-C and SN-F can access each other by SSH only. SSH access from your home address is allowed for administration reasons.

SUBMIT for task 2b

✓ The snippets of your solution (what you used Security Group or NACLs or combination).
✓ The explanation of your solution.

- Implement the following case by customizing the **NACL**s or the **Security Group**s or both.



- The machine VM3 and VM6 are Linux Ubuntu.
- VM3 and VM6 can be accessed from your home address with SSH and HTTP.
- VM3 attached to subnet SN-C must be able to SSH into VM6 and vice versa.
- The previous conditions are exclusive. No other traffic, from anywhere is allowed.

SUBMIT for task 2c

✓ The snippets of your solution (what you used, Security Group or NACLs or a combination).
✓ The explanation of your solution.

TASK 3: Creation of EC2 instances

- Create the following:
- Make sure to include your name in the VMs (example VM1-saber)

EC2-instance	Home Subnet	AMI	Instance Type
VM1	SN-A	Amazon CentOS default	T2 micro Free Tier
VM2	SN-B	Amazon CentOS default	T2 micro Free Tier
VM3	SN-C	Linux Ubuntu Server 20.04	T2 micro Free Tier
VM4	SN-D	Windows Server 2019	T2 micro Free Tier
VM5	SN-E	Windows Server 2019	T2 micro Free Tier
VM6	SN-F	Linux Ubuntu Server 20.04	T2 micro Free Tier

- Note: make sure that the rules allow the machines to download the packages necessary to do bootstrapping.
- Note: make sure that your EC2 instances land in the specified subnets.

- For the following instances, bootstrap the installation to make them web servers (test that they work).

EC2-instance	Application
VM1	httpd (Apache2)
VM2	httpd (Apache2)
VM3	NGINX
VM6	NGINX

SUBMIT for Task 3:

- ✓ The SSH to all the Linux based machines (that include the SSH between the EC2s).
- ✓ Submit the snippets of the prompts with the output of either the **ifconfig** (CentOS) or **ip addr (Ubuntu)** command.
- ✓ Remote Desktop Connection (RDP) to all the Windows Servers.
- ✓ Submit the snippets of the successful access.
- ✓ The snippets of the web browser to each webserver.

Submission:

- ✓ Submit one document report in Word or PDF.
- ✓ Make sure that all relevant screenshot for each task is clearly laid out.
- ✓ Please see marking scheme posted on course shell.
- ✓ Assignment 1 is worth 10% of your final grade.